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Introduction:

The Smart Parking Simulator is a hardware-based embedded systems project developed as part of the Digital Logic Design. The project demonstrates the practical application of combinational logic circuits in solving a real-world problem: automated parking lot.

The core design principle is that all logical decision-making regarding parking availability is performed entirely by hardware logic gate ICs (**7404 NOT gate and 7408 AND gate**). The Arduino Nano microcontroller acts only as a signal relay and actuator driver, making no logical decisions in software.

The system monitors 4 parking slots using HW-201 IR sensors, detects approaching vehicles using an HC-SR04 ultrasonic sensor, and controls an SG90 servo motor barrier along with a 5V active buzzer to manage parking entry.

Objectives:

The primary objectives of this project are:

- To detect slot occupancy using HW-201 IR infrared sensors (active-LOW).
- To relay sensor signals through Arduino Nano to external hardware logic ICs without any software decision-making.
- To use the 7404 NOT gate IC to invert active-LOW sensor signals to active-HIGH logic levels.
- To use the 7408 AND gate IC to compute $\text{FULL} = S1 \cdot S2 \cdot S3 \cdot S4$ entirely in hardware.
- To detect approaching vehicles at the parking entry gate using HC-SR04 ultrasonic sensor.



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- To control an SG90 servo motor barrier that opens only when space is available and a car is detected.
- To activate a buzzer alert only when parking is full AND a car is detected at the gate simultaneously.

Problem Statement:

Urban parking management presents a growing challenge. Drivers waste time searching for available spaces, contributing to traffic congestion and fuel wastage. Manual systems are inefficient while fully computerized systems are expensive.

This project solves the following specific problems:

- Inability to automatically detect and communicate slot occupancy in real time.
- Lack of a gate control mechanism preventing entry when the lot is full.
- Absence of an audio alert system to notify drivers of a full lot before approaching the barrier.
- The challenge of implementing all decision logic in hardware gates, demonstrating DLD principles in a real functional system.

System Overview:

Block	Components	Role
Sensor Block	4x HW-201 IR sensors + HC-SR04	Detect slot occupancy and vehicles at gate
Signal Relay	Arduino Nano	Read sensors, forward signals, drive actuators
Logic Block	7404 NOT gate + 7408 AND gate	Compute FULL signal entirely in hardware
Output Block	SG90 servo motor + 5V buzzer	Physical barrier control and audio alert
Power Block	Stable 5v USB port	Supply regulated 5V to all components



Components and Hardware:

#	Component	Model	Purpose
1	Arduino Nano	ATmega328P	Signal relay, sensor interface
2	IR Sensor Module	HW-201, Active-LOW, 5V	Detect slot occupancy (one per slot)
3	Ultrasonic Sensor	HC-SR04	Vehicle detection at entry gate
4	NOT Gate IC	7404, 14-pin	Invert LOW signals to HIGH
5	AND Gate IC	7408, 14-pin	Compute FULL signal in hardware
6	Servo Motor	SG90, 5V	Physical barrier
7	Active Buzzer	5V active buzzer	Alert driver when parking is full
8	Breadboard	Full-size	Circuit assembly platform
9	Jumper Wires	Male-male, male-female	All component interconnections

Working Principle:

➤ IR Sensor Operation

Each parking slot has one HW-201 IR sensor. The HW-201 operates on the active-LOW principle:

- Car IS present in slot → sensor output = LOW (0V)
- Slot IS empty → sensor output = HIGH (5V)

The Arduino reads these on A0, A1, A2, A3 and forwards the raw uninverted readings to the 7404 IC from D2, D3, D4, D5.

➤ 7404 NOT Gate IC

The 7404 contains 6 independent NOT gates. This project uses 5 of them:



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Gate	Input Pin	From	Output Pin	Output Signal
Gate 1	Pin 1 (1A)	Arduino D2	Pin 2	$S1' = \text{inverted } S1$
Gate 2	Pin 3 (2A)	Arduino D3	Pin 4	$S2' = \text{inverted } S2$
Gate 3	Pin 5 (3A)	Arduino D4	Pin 6	$S3' = \text{inverted } S3$
Gate 4	Pin 9 (4A)	Arduino D5	Pin 8	$S4' = \text{inverted } S4$

➤ **7408 AND Gate IC**

The 7408 contains 4 AND gates. Gates 1, 2, 3 are chained to compute FULL.
Gate 4 computes the entry OPEN condition.

Gate	Input A	Input B	Output	Meaning
Gate 1	$S1'$ from 7404 pin 2	$S2'$ from 7404 pin 4	$S1' * S2'$	Slots 1 and 2 both full
Gate 2	$S3'$ from 7404 pin 6	$S4'$ from 7404 pin 8	$S3' * S4'$	Slots 3 and 4 both full
Gate 3	Gate 1 output-pin 3	Gate 2 output-pin 6	FULL	ALL 4 slots occupied

Boolean expression

$$\text{FULL} = S1' * S2' * S3' * S4'$$

➤ **HC-SR04 Entry Gate Detection**

Positioned at the parking entry gate, the HC-SR04 sends a 10 micro-second trigger pulse on D8 and measures the echo on D7

If distance is less than 20 cm, a car is detected. Arduino sets D6 HIGH to signal this to 7408 Gate 4.



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➤ **Barrier and Buzzer Behavior**

The system operates across four distinct cases

Parking Full?	Car at Gate?	Barrier	Buzzer
YES	YES	CLOSED	ON
YES	NO	CLOSED	OFF
NO	YES	OPEN (3s)	OFF
NO	NO	CLOSED	OFF

Truth Table:

All 16 input combinations for the 4 parking slots

S1	S2	S3	S4	S1'	S2'	S3'	S4'	FULL	Barrier	Buzzer
1	1	1	1	0	0	0	0	0	Open	OFF
0	1	1	1	1	0	0	0	0	Open	OFF
1	0	1	1	0	1	0	0	0	Open	OFF
0	0	1	1	1	1	0	0	0	Open	OFF
1	1	0	1	0	0	1	0	0	Open	OFF
0	1	0	1	1	0	1	0	0	Open	OFF
1	0	0	1	0	1	1	0	0	Open	OFF
0	0	0	1	1	1	1	0	0	Open	OFF
1	1	1	0	0	0	0	1	0	Open	OFF
0	1	1	0	1	0	0	1	0	Open	OFF
1	0	1	0	0	1	0	1	0	Open	OFF
0	0	1	0	1	1	0	1	0	Open	OFF
1	1	0	0	0	0	1	1	0	Open	OFF
0	1	0	0	1	0	1	1	0	Open	OFF
1	0	0	0	0	1	1	1	0	Open	OFF
0	0	0	0	1	1	1	1	1	Close	ON



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Complete Pin Connection:

➤ Arduino Nano Left Side Pins (D2 to GND)

Arduino Pin	Direction	Connects To	Purpose
D2	OUTPUT	7404 Pin 1 (1A)	signal S1 NOT gate
D3	OUTPUT	7404 Pin 3 (2A)	signal S2 NOT gate
D4	OUTPUT	7404 Pin 5 (3A)	signal S3 NOT gate
D5	OUTPUT	7404 Pin 9 (4A)	signal S4 NOT gate
D7	INPUT	HC-SR04 ECHO	Receive echo pulse
D8	OUTPUT	HC-SR04 TRIG	Send trigger pulse
D9	OUTPUT	Servo orange wire	PWM
D10	OUTPUT	Buzzer +	HIGH = buzzer ON
D11	INPUT	7408 Pin 8	FULL signal AND
5V	POWER	5V breadboard rail	Supply 5V
GND	POWER	GND rail	Common ground

➤ Arduino Nano Right Side Pins

Arduino Pin	Direction	Connects To	Purpose
A0	INPUT	IR Sensor 1 OUT	Slot 1 state
A1	INPUT	IR Sensor 2 OUT	Slot 2 state
A2	INPUT	IR Sensor 3 OUT	Slot 3 state
A3	INPUT	IR Sensor 4 OUT	Slot 4 state
VIN (optional)	POWER	9V +	9V in terminal
GND	POWER	9V GND	Battery negative



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➤ **7404 NOT Gate IC**

Pin	Name	Direction	Connects to	Purpose
Pin 14	VCC	POWER	5V breadboard rail	Power supply for IC
Pin 7	GND	POWER	GND rail	Ground for IC
Pin 1	1A	INPUT	Arduino D2	Receives S1 raw signal
Pin 2	1Y	OUTPUT	7408 Pin 1 (1A)	$S1' \rightarrow$ AND gate input
Pin 3	A2	INPUT	Arduino D3	Receives S2 raw signal
Pin 4	2Y	OUTPUT	7408 Pin 2 (1B)	$S2' \rightarrow$ AND gate input
Pin 5	3A	INPUT	Arduino D4	Receives S3 raw signal
Pin 6	3Y	OUTPUT	7408 Pin 4 (2A)	$S3' \rightarrow$ AND gate input
Pin 9	4A	INPUT	Arduino D5	Receives S4 raw signal
Pin 8	4Y	OUTPUT	7408 Pin 5 (2B)	$S4' \rightarrow$ AND gate input

➤ **7408 AND Gate IC**

Pin	Name	Direction	Connects to	Purpose
Pin 14	VCC	POWER	5V breadboard rail	Power supply for IC
Pin 7	GND	POWER	GND rail	Ground for IC
Pin 1	1A	INPUT	7404 Pin 2 (1Y)	Receives S1'
Pin 2	1B	INPUT	7404 Pin 4 (2Y)	Receives S2'
Pin 3	1Y	OUTPUT	7408 Pin 9 (3B)	$S1' * S2'$ result
Pin 4	2A	INPUT	7404 Pin 6 (3Y)	Receives S3'
Pin 5	2B	INPUT	7404 Pin 8 (4Y)	Receives S4'
Pin 6	2Y	OUTPUT	7408 Pin 10 (3A)	$S3' * S4'$ result
Pin 10	3A	INPUT	7408 Pin 6(internal)	Receives $S3' * S4'$
Pin 9	3B	INPUT	7408 Pin 3(internal)	Receives $S1' * S2'$
Pin 8	3Y	OUTPUT	Arduino D6	FULL signal: HIGH



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➤ **IR Sensors, HC-SR04, Servo, Buzzer, Battery**

Component	Pin/Wire	Connects To	Purpose
IR Sensor 1	VCC	5V rail	Power
IR Sensor 1	GND	GND rail	Ground
IR Sensor 1	OUT	Arduino A0	Slot 1 occupancy signal
IR Sensor 2	VCC	5V rail	Power
IR Sensor 2	GND	GND rail	Ground
IR Sensor 2	OUT	Arduino A1	Slot 2 occupancy signal
IR Sensor 3	VCC	5V rail	Power
IR Sensor 3	GND	GND rail	Ground
IR Sensor 3	OUT	Arduino A2	Slot 3 occupancy signal
IR Sensor 4	VCC	5V rail	Power
IR Sensor 4	GND	GND rail	Ground
IR Sensor 4	OUT	Arduino A3	Slot 4 occupancy signal
HC-SR04	VCC	5V rail	Power
HC-SR04	GND	GND rail	Ground
HC-SR04	TRIG	Arduino D8	Arduino sends trigger pulse
HC-SR04	ECHO	Arduino D7	Sensor returns echo pulse
SG90 Servo	VCC	5V rail	Power
SG90 Servo	GND	GND rail	Ground
SG90 Servo	PWM	Arduino D9	PWM control signal
Buzzer	VCC	Arduino D10	Signal: HIGH = ON
Buzzer	GND	GND rail	Ground



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System Behavior and Test Cases:

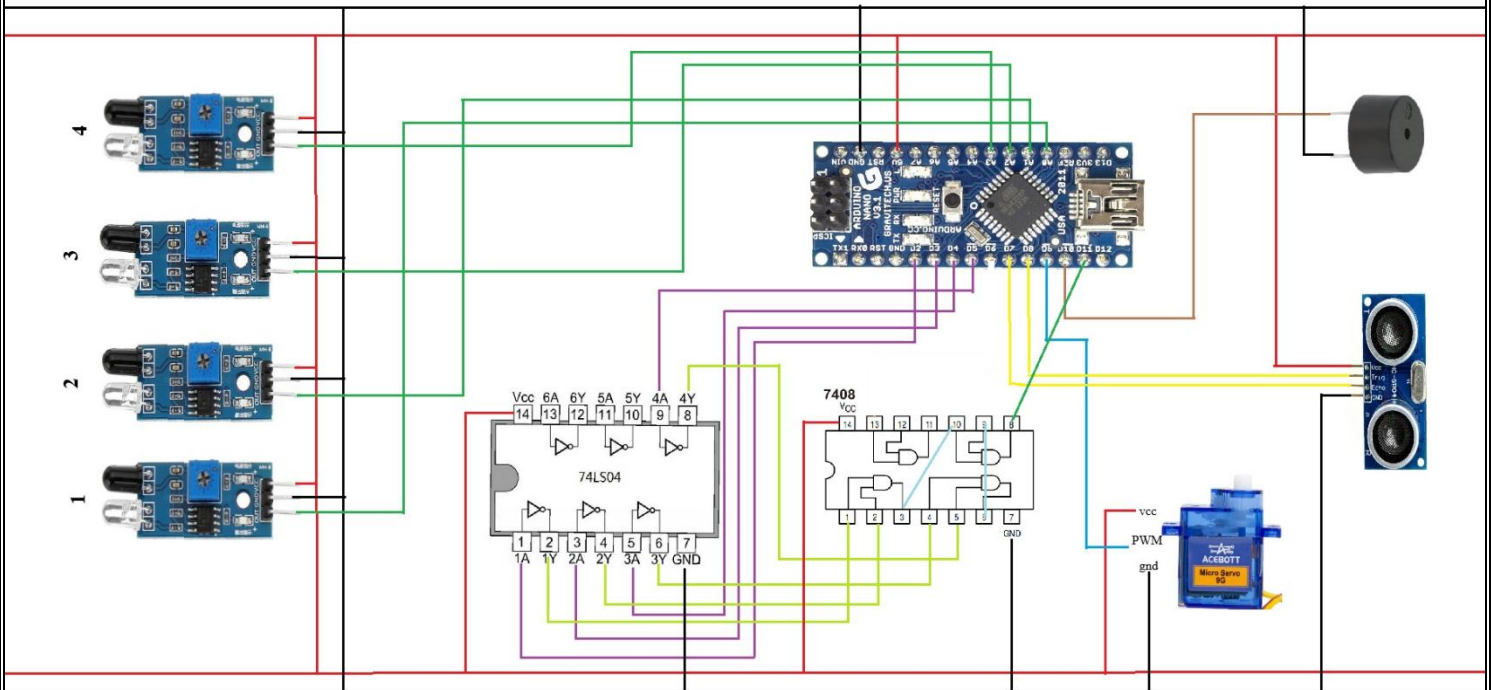
Occupied	Car at Gate?	7408 FULL	Barrier	Buzzer	Result
0 of 4	No	LOW (0)	Closed	OFF	Idle, waiting
2 of 4	Yes	LOW (0)	Opens 3s	OFF	Car enters normally
3 of 4	Yes	LOW (0)	Opens 3s	OFF	Car enters normally
4 of 4	Yes	HIGH (1)	Closed	OFF	Full but silent
4 of 4	Yes	HIGH (1)	Closed	ON	Full + car: buzzer
4 to 3	Yes	LOW (0)	Opens 3s	OFF	Car left slot

Applications:

- **Shopping malls and commercial parking lots:** automatically prevent entry when full.
- **Hospital parking:** priority access management for emergency vehicles.
- **University campus:** monitor limited faculty and student parking zones.
- **Airport short-term parking:** manage high-turnover slot availability.
- **Smart city infrastructure:** scalable to IoT-based city-wide parking guidance systems.
- **Industrial automation:** the same combinational logic applies to inventory and conveyor slot tracking.



Circuit Diagram:



Limitations:

- Fixed slot count: The current design supports exactly 4 parking slots. Adding more slots would require additional NOT gate and AND gate ICs.
- No display output: The system has no LCD or 7-segment display showing the **number of available slots**. Availability is only indicated indirectly through barrier and buzzer behavior.
- Single entry gate: Only one entry gate is supported. A real parking lot with multiple entry and exit gates would require a significantly more complex circuit.
- No exit detection: The system detects when slots are occupied but has no dedicated exit gate. When a car leaves a slot, the IR sensor simply detects the vacancy and the FULL signal drops automatically.
- Servo as barrier: The SG90 servo is a small model servo not suitable for actual vehicle barriers. A real implementation would require a heavy-duty motor or solenoid actuator.



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- **IR sensor interference:** HW-201 IR sensors can be affected by bright ambient light (especially sunlight), which may cause false detections outdoors.
- **3-second fixed delay:** The barrier open time is fixed at 3 seconds regardless of vehicle speed. A slow vehicle may not clear the gate in time.

Future Improvements:

- **LCD or LED display:** Add a 16x2 LCD display or 7-segment display (using 7447 BCD-to-7-segment IC, already available in the lab) to show the number of available slots in real time.
- **IoT integration:** Add an ESP8266 or ESP32 WiFi module to transmit parking availability data to a mobile app or web dashboard, enabling drivers to check availability before arriving.
- **Exit gate sensor:** Add a separate HC-SR04 or IR sensor at the exit to detect departing vehicles and automatically decrement the occupied slot count.
- **RFID access control:** Integrate an RFID reader at the entry gate so that only authorized vehicles (with registered RFID cards) can enter regardless of space availability.
- **More parking slots:** Extend the logic to 8 or 16 slots by adding additional 7404 and 7408 ICs in cascade. Each additional stage doubles the slot capacity.
- **Solar power:** Replace the 9V battery with a small solar panel and rechargeable Li-ion battery pack for outdoor installation without access to wall power.
- **Real barrier motor:** Replace the SG90 servo with a high-torque DC motor or linear actuator suitable for controlling a physical boom barrier in a real parking lot.
- **Automated slot guiding LEDs:** Add a green LED for each empty slot and red LED for each occupied slot, visible from the car entrance, to guide drivers to available spaces.
- **Vehicle counting:** Add a counter using a JK flip-flop (74LS76, available in lab) to track total vehicles entered and exited for daily reporting.



Conclusion:

The Smart Parking Simulator successfully demonstrates the application of Digital Logic Design principles in a real, functional embedded system. The project achieves its core objective: all parking-full decision logic is implemented in hardware using the 7404 NOT gate IC and 7408 AND gate IC, with the Arduino Nano serving strictly as a signal relay and actuator driver.

The Boolean expression $FULL = S1' * S2' * S3' * S4'$ is computed entirely by the 7408 AND gate chain in hardware. The buzzer activates only when the parking lot is full AND a vehicle is simultaneously detected at the entry gate, avoiding unnecessary noise when the lot is full but no one is approaching.

The project reinforces the fundamental principle that digital logic design forms the backbone of every modern automated and embedded system, demonstrating that combinational circuits can implement intelligent system behavior without relying on software decision-making.